

SANJAY SATHYAMOORTHY

Chennai, Tamil Nadu | 9385619541 | gss.sanjay23@gmail.com

linkedin.com/in/sanjay-s-6b48a6200 | github.com/sanjay-923 | sanjay-923.github.io/sanjay-923

PROFESSIONAL SUMMARY

Electronics and hardware engineer with hands-on experience in PCB design, RS485 Modbus RTU network architecture, and harsh-environment sensor integration for agricultural IoT automation. Promoted from intern to Associate Electronics Engineer at Akaththi Farms within 6 months, delivering multi-node automation hardware across sensor, actuator, power monitoring, and HMI subsystems. B.E. Electronics & Communication Engineering, Rajalakshmi Institute of Technology.

TECHNICAL SKILLS

EDA Tools	KiCad 8 (primary), Altium Designer (coursework), Proteus 8
Protocols	RS485, Modbus RTU, I ² C, UART, RS232, MQTT
Hardware Platforms	ESP32, ESP32-S3, ESP8266, Xilinx Spartan-6 FPGA (LX9)
PCB & Hardware	Schematic capture, PCB layout, DRC, BOM management, Gerber file generation, hand soldering, SMD rework
Design Practices	Harsh-environment component qualification (IP rating, humidity tolerance), datasheet-driven component selection, JST modular connector design, ADR documentation
Hardware Validation	Multimeter, bench testing, UART terminal monitoring
Programming	Python, Embedded C
Tools	Docklight RS232/UART terminal, Git
Languages	Tamil (native), English (professional)

PROFESSIONAL EXPERIENCE

Akaththi Farms <i>Associate Electronics Engineer</i>	July 2025 – Present <i>Chennai, Tamil Nadu</i>
- Architected a 4-node industrial automation system for mushroom cultivation pods — sensor, actuator, power monitoring, and HMI nodes — communicating over RS485 Modbus RTU with WiFi for inter-pod connectivity. - Diagnosed I²C signal degradation on long intra-pod cable runs and re-architected communication to RS485 Modbus RTU, enabling a reliable addressable multi-drop bus topology across all node types. - Designed KiCad 8 PCB layouts for all 4 node types, defining pin mapping, power constraints, Modbus RTU slave addressing, and Gerber files for fabrication. - Specified the Waveshare ESP32-S3 7-inch touch LCD as the local HMI node, enabling on-site real-time monitoring and control of pod parameters via an RS485-connected local dashboard. - Qualified all sensor and actuator components for 95% RH operation through IP rating analysis and humidity tolerance review; standardized JST connectors for tool-free field replacement.	
Akaththi Farms <i>Electronics Engineering Intern</i>	January 2025 – July 2025 <i>Chennai, Tamil Nadu</i>
- Initiated hardware prototyping for mushroom pod automation, delivering v1 (exhaust fan timer control) and v2 (CO₂-threshold override with temperature, humidity, and lux monitoring) across 9 hand-soldered prototype boards. - Evaluated component datasheets across CO₂, temperature/humidity, lux, and water-level sensing categories for agricultural IoT deployment and prepared ADR documentation for all selection decisions. - Designed circuit architecture and pin mapping for sensor, actuator, power grid, and HMI nodes, considering power constraints and communication interface requirements per node type.	

- Designed PCB layout for an FPGA-based humidity and temperature sensor system for cyclotron gas detector qualification using a Xilinx Spartan-6 FPGA (LX9); hand-soldered SHT21 bare-die SMD sensor directly onto the board.
- Implemented UART communication via Docklight RS232 terminal and wrote Python scripts to process and visualize real-time sensor output for accuracy analysis.
- Validated system reliability through an uninterrupted 48-hour continuous run, achieving 99% measurement accuracy for particle physics instrumentation.

ACADEMIC PROJECTS

Inverter Monitor and Control Using IoT

Final Year Project – 2024

B.E. Electronics & Communication Engineering, Rajalakshmi Institute of Technology

- Designed PCB layout for an ESP32-based inverter system with ESP8266 slave nodes, lithium-ion battery, and ferrite core transformer using Proteus 8.
- Integrated MQTT protocol for IoT-based remote monitoring, enabling real-time parameter adjustment and operational status visibility.

EDUCATION

Bachelor of Engineering — Electronics and Communication Engineering

Rajalakshmi Institute of Technology, Chennai

Graduated May 2024

CGPA: 8.25 / 10

Higher Secondary Certificate (HSC)

Shanthinikethan Matriculation Higher Secondary School, Vellore

March 2020

Percentage: 67.8%

Secondary School Leaving Certificate (SSLC)

Shanthinikethan Matriculation Higher Secondary School, Vellore

March 2018

Percentage: 82.8%

CERTIFICATIONS

- PCB Design with Altium Designer — Udemy
- PCB Design with KiCad — Udemy

AWARDS & RECOGNITION

- **Monetary Prize — AAVISHKAAR Technical Symposium**, Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College — recognized for technical project presentation.
- **Paper Presentation — ICRTICPT 2022 International Conference** — presented on Internet of Things applications to 200+ professionals and academics.
- **FPGA Training Workshop Facilitator** — led a 2-day Quartus Prime Lite workshop, training team members in FPGA programming and device configuration.
- **Industrial Visit — Flex Electronics** — observed advanced PCB assembly and SMT manufacturing processes.